ARM PrimeCell
STATIC MEMORY CONTROLLER (PL092)
Errata Notice

PL092 DEI 0000 1.4

ARM PrimeCell STATIC MEMORY CONTROLLER PL092 Errata Notice

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General suggestion for additions and improvements are also welcome.

Contents

1 ABOUT THIS DOCUMENT

1.1 Change control

1.1.1 Change history

This refers to the revision changes to the Errata document

Issue	Date	Change
1.0	12th November 02	First Draft
1.1	12 December 02	Updated with review comments
1.2	1 Feb. 2003	Updated for release r1p2-00ltd0
1.3	20th May 2003	Updated for r1p3-00ltd0
1.4	4th September 2003	Updated for r1p3-01ltd0

1.2 Errata List

This table summarises the errata notices detailed

Release: Release revision number affected

Section: Section in the document where the issues are described

Status: The status conditions are:

Open: Issue is current

Closed: Issue resolved in subsequent release

Severity: The severity conditions are:

A **Low** level of severity indicates that there is no direct impact to the silicon implementation. Examples are:

- No change to RTL required
- Documentation change
- Test pattern change
- Synthesis Scripts change

A **Medium** level of severity indicates that there is a minor impact to the silicon implementation, which can be overcome by modifications detailed in the errata notes. Examples are:

- A specific sequence of events not adhered to.
- Ambiguous or unexpected operation of the device.

A **High** level of severity indicates that there is a major impact to the silicon implementation, which cannot be overcome without modifications detailed in the errata notes. Examples are:

- Non conformance to the device specification.

Affected: Deliverables impacted by the issue

Description: Issue concerning Errata

Release	Section	Status	Severity	Affected	Description
1v1	2.1	Closed	Low	RTL	HREADY signal driven HIGH prematurely in AddrToggleTests.c
1v1	2.2	Closed	High	RTL	Deadlock occurs when there is a change in memory configuration
1v1	2.3	Closed	High	RTL	SMC generates incorrect accesses when using different sized banks
1v1	2.4	Closed	High	RTL	A write following a burst transfer is ignored when no wait states are used
1v1	2.5	Closed	Low	RTL	Booting is not possible from 16/32-bit devices with Byte Lane Inputs on Chip Select 7
1v1	2.6	Closed	Medium	RTL	Issues relating to SMWAIT Signal
1v1	2.7	Closed	Medium	RTL	Issues with DBI/EBI
1v1	2.8	Closed	Medium	RTL	Error in nSMDATAEN generation logic
1v1	2.9	Closed	Medium	RTL	Error in IDLE cycle generation logic
1v1	2.10	Closed	Medium	RTL	Error occurs in AHB burst writes where HSIZE is less than memory width
1v1	2.11	Closed	Low	RTL	HREADYOUT generated prematurely
1v1	2.12	Closed	Low	RTL	Register write occurs with zero AHB wait states

1v1	2.13	Closed	Medium	RTL	A memory transfer takes place during an IDLE cycle with Burst Mode devices
1v1	2.14	Closed	High	RTL	nSMCBLSOUT and nSMCS may be incorrectly generated
1v1	2.15	Closed	High	RTL	SMCDATAOUT is generated one cycle late for memory writes with AHB size (HSIZE) greater than the memory size (MSIZE)
1v1	2.16	Closed	Medium	RTL	Issues with AHB WRAP4 transfers and end of burst internal signals
1v1	2.17	Closed	Low	RTL	Clarification of WSTOEN and WSTWEN programmed values
1v1	2.18	Closed	High	RTL	Transfer state machine signal not deasserted at end of Burst Mode Read
1v1	2.19	Closed	Low	RTL	Burst Mode indicator incorrectly deasserted at the end of the first read of two read transfers
r1p2	3.1	Closed	Low	RTL	Booting is not possible from 16/32-bit devices with Byte Lane inputs on chip select 7
r1p2	3.2	Closed	Medium	RTL	Issues with SMWAIT signal
r1p2	3.3	Closed	Medium	RTL	Issues with DBI/EBI
r1p2	3.4	Closed	Medium	RTL	Double assignment to same signal in SmcAhbif.vhd
r1p3	4.1	Closed	Low	Trickbox	SmcTM30 timing check is incorrect - and timing violations README is missing
r1p3	4.2	Closed	High	DBI.v	SmcState Machine Locks in ST_BUFWR state

2 PL092 RELEASE 1V1 - ERRATA DETAILS

2.1 HREADY Signal Driven HIGH Prematurely in AddrToggleTests.c

2.1.1 Summary

In running the AddrToggleTests.c tests of the PL092 BusTalk test-bench: HREADY signal was asserted HIGH in error before the access was completed. This occurred during the first transaction to Chip Select 1, at the same time as reset was occurring.

2.1.2 Severity

Low – this scenario is expected to be rare in practice

2.1.3 Symptoms

The conditions under which this behaviour would occur are as follows. Both of them need to occur to hit this particular issue.

-> A transfer is started, and on the same clock edge, HRESETn is being de-asserted.

-> A transition is happening from HSELSMC to HSELREG on the second clock edge.

This means a burst started on the same clock cycle when the reset is being de-asserted. The burst is broken in the second clock cycle; when an access is being started to the register interface (in the same clock cycle). In any other place if this condition occurs, this failure does not occur.

2.1.4 Affected Items

RTL.

2.1.5 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.2 Deadlock Occurs When There is a Change in Memory Configuration

2.2.1 Summary

The SMC has difficulty in recognising changes to its configuration in some circumstances and so locks up.

2.2.2 Severity

High

2.2.3 Symptoms

This issue occurs in the SMC when:

- A BUSY transfer is seen during HSELSMC low or
- HREADYIN low when the SMC is carrying out a memory write or
- In an access to a different AHB slave during a write burst.

2.2.4 Affected Items

RTL

2.2.5 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.3 SMC Generates Incorrect Accesses When Using Different Sized Banks

2.3.1 Summary

Memory width information is not being updated internally for an access following a burst write to a different chip select.

2.3.2 Severity

High

2.3.3 Symptoms

This bug may be seen in the following ways:

- A WORD read from 32-bit memory width bank fails immediately after a BYTE write to a 16-bit memory width bank.

- An incorrect access may also be seen when an odd number of writes with HSIZE less than memory size is followed by a read or write to a different chip select.

This may be coupled with the SMC AHB accesses being locked by HREADYOUT being taken LOW.

The occurrence of this condition sets an internal signal erroneously and affects the subsequent memory accesses.

2.3.4 Affected Items

RTL

2.3.5 Short Term Workaround

2.3.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.4 A Write Following a Burst Transfer is Ignored When No Wait States Are Used

2.4.1 Summary

There is an issue whereby a write following a burst transfer is missed when there are no wait states.

2.4.2 Severity

High

2.4.3 Symptoms

Same as summary.

2.4.4 Short Term Workaround

A software workaround is to add wait states to the existing code

2.4.5 Affected Items

RTL

2.4.6 Proposed Corrective action

The RTL has been fixed in version r1p2-00ltd0.

2.5 Booting is not possible from 16/32-bit devices with Byte Lane Inputs on Chip Select 7

2.5.1 Summary

Tie-off inputs are provided to set the size of the memory connected to chip select 7, but no tie-off is provided to set the RBLE bit of the bank control register for chip select 7, which by default is set for 8-bit devices.

2.5.2 Severity

Low

Booting possible from 8-bit devices

2.5.3 Symptoms

Unable to read from 16-bit or 32-bit memory devices with byte lane inputs on chip select 7 immediately after reset. The nSMBLS outputs are only driven during write transfers when the RBLE bit of the bank control register is set to zero (8-bit device mode).

2.5.4 Short Term Workaround

Use 8-bit memory devices for booting, or 16/32-bit devices without byte lane inputs.

2.5.5 Affected Items

RTL

2.5.6 Proposed Corrective action

Add a tie-off input for the chip select 7 RBLE bit to allow configuration along with the memory size.

The issue has been fixed in version r1p3-00ltd0.

2.6 Issues Relating to SMWAIT Signal

2.6.1 Summary

If a number of transfers are performed to a number of different chip selects with different external wait enabled settlings (selecting the use of the SMWAIT input to control the timing of the transfer instead of the WST1 and WST2 registers) or to a single chip select with the external wait enabled, the turnaround time between the transfers may be incorrectly generated (the counter is loaded with the programmed turnaround value, but is then cleared in the following cycle, and the SMC may lock up by holding HREADYOUT LOW.

2.6.2 Severity

Medium

This issue will only occur if the SMWAIT external wait input is used to control the timing of the transfers on any chip select.

2.6.3 Symptoms

As summary

2.6.4 Short Term Workaround

If possible, do not use the SMWAIT signal for any transfer, but use only the WST1 and WST2 registers to control the timing of the transfers.

2.6.5 Affected Items

RTL.

2.6.6 Proposed Corrective action

The issue has been fixed in version r1p3-00ltd0.

2.7 Issues With DBI/EBI

2.7.1 Summary

1) The SMC may incorrectly drive the external bus even when the grant input is not asserted to indicate that the SMC has control of the external memory bus. The grant may be from either the internal DBI or the external EBI, depending on the setting of the EXTBUSMUX input. This means that some transfers

may be performed where the control signals are driven by the SMC and the address and data signals are driven by the device using either the MC port or the currently granted port of the EBI.

2) If the external memory bus grant is deasserted during memory accesses, subsequent transfers may be incorrectly generated. The grant may be from either the internal DBI or the external EBI, depending on the setting of the EXTBUSMUX input. Some cases may cause HREADYOUT to stay LOW locking up the AHB, and others may cause data transfers to be performed incorrectly, using the control signals from the SMC and address and data from the device which is granted control of the bus.

2.7.2 Severity

Medium

Both issues will only occur if another device is used to drive the external memory interface through the DBI or the EBI. If the SMC is the only device using the external bus then it will always be granted control of the external bus.

2.7.3 Symptoms

As summary

2.7.4 Short Term Workaround

If possible the EBI should not be used and the SMC should be the only device driving the external interface.

2.7.5 Affected Items

RTL.

2.7.6 Proposed Corrective action

The issue has been fixed in version r1p3-00ltd0.

2.8 Error in nSMDATAEN Generation Logic

2.8.1 Summary

An error in the nSMDATAEN generation logic in the DBI means that if the ‘MC’ port is granted control of the bus, nSMDATAEN could be driven with the value of nSMCDATAEN from the SMC instead of from the MCDATAEN input.

2.8.2 Severity

Medium

This issue only occurs if the ‘MC’ port is used.

2.8.3 Symptoms

As Summary.

2.8.4 Short Term Workaround

If possible the 'MC' port should not be used.

2.8.5 Affected Items

RTL.

2.8.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.9 Error In IDLE cycle Generation Logic

2.9.1 Summary

An error in the IDLE cycle generation logic means that if accesses are performed to changing chip selects, the programmed IDCY value for the current transfer's chip select may not be correctly loaded into the counter, and a value from a different chip select may be used instead.

2.9.2 Severity

Medium

This issue does not cause a problem if the IDCY value is programmed to be the same for all chip selects in use.

2.9.3 Symptoms

For example, if the first access is to chip select 1 and the second access is to chip select 2, chip select 1's transfer may proceed with chip select 2's IDCY programmed value.

2.9.4 Short Term Workaround

Program the IDCY value to be the same for all chip selects in use

2.9.5 Affected Items

RTL.

2.9.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.10 Error Occurs in AHB Burst Writes where HSIZE is Less Than Memory Width

2.10.1 Summary

An error occurs during an AHB burst of write transfers, where the AHB transfer size (HSIZE) is less than the memory width, and the burst is broken (the AHB Arbiter de-asserts the HBUSGRANT signal to the AHB master currently performing the burst of writes) while the master is performing a BUSY transfer.

2.10.2 Severity

Medium

This issue does not occur if BUSY transfers are never performed by the AHB master, or if the AHB arbiter never breaks a burst.

2.10.3 Symptoms

This issue only occurs if the AHB master performs busy transfers and the AHB arbiter breaks the AHB bursts.

The normal operation of the AHB arbiter is to change the currently granted master after the current AHB burst has ended.

2.10.4 Affected Items

RTL

2.10.5 Short Term Workaround

If an AHB master is used which generates BUSY transfers, ensure that the AHB arbiter will never break a burst.

2.10.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.11 HREADYOUT Generated Prematurely

2.11.1 Summary

This issue may be related to Issue 2.1.

The HREADYOUT output is generated incorrectly if a register read is performed on the first rising edge of HCLK after the reset has been deasserted (HRESETn=1), and the second transfer performed is to a memory location, with HTRANS either IDLE or NSEQ, and the transfer is either a read or write (HWRITE = 0 or 1).

2.11.2 Severity

Low

In a standard ARM CPU-based system, the first AHB transfer after reset will be an instruction fetch from memory, so this issue will not occur as a register read is never performed until after the first cycle after reset.

2.11.3 Symptoms

HREADYOUT should be held LOW for 2 cycles, one for internal initialization and once cycle for the register read; but HREADYOUT is only held LOW for one cycle.

The read data is correctly generated after two cycles, but will not be read at the correct time due to the early HREADYOUT generation.

2.11.4 Short Term Workaround

This issue does not occur if the first register read is performed on the second or subsequent cycles after the reset is deasserted.

2.11.5 Affected Items

RTL

2.11.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.12 Register Write Occurs with Zero AHB Wait States

2.12.1 Summary

If a write to memory is immediately followed by a write to a register, the register write will incorrectly receive zero AHB wait states. The correct operation is for all register writes to receive one AHB wait state (HREADYOUT LOW for one HCLK cycle).

As one cycle is required for the newly written register values to propagate through to the memory control logic, the memory transfer performed after the zero wait register write may use the previous register values, rather than the new register values as the memory access will be started one cycle early.

2.12.2 Severity

Low

This issue will not occur if a register write is only performed after a register read (Read-Modify-Write sequence).

2.12.3 Symptoms

As Summary

2.12.4 Short Term Workaround

1) This issue will not occur if a register write is only performed after a register read (Read-Modify-Write sequence).

2) A software workaround for this issue is to ensure that a memory access will never be performed immediately after a register write. This may be achieved by always:

- performing a register read after a register write,

- inserting an IDLE cycle, or

- accessing another device.

2.12.5 Affected Items

RTL

2.12.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.13 A Memory Transfer Takes Place During an IDLE Cycle With Burst Mode Devices

2.13.1 Summary

If an IDLE write transfer is generated at the end of an external burst mode memory read (BM=1), the memory write transfer will be performed.

2.13.2 Severity

Medium

This issue is not a problem if Burst Mode memories are not used.

2.13.3 Symptoms

As the transfer is an IDLE, no memory write should be performed, so data corruption may occur, depending on the address and data value of the transfer.

2.13.4 Short Term Workaround

This issue will not occur if all AHB IDLE transfers drive HWRITE=0 (Read transfer), or if no Burst Mode memories are used (BM=0).

2.13.5 Affected Items

RTL.

2.13.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.14 nSMCBLSOUT And nSMCS May Be Incorrectly Generated

2.14.1 Summary

nSMCBLSOUT and nSMCS may be incorrectly generated when a write to memory is followed by a second write to memory. The issue only occurs when the second AHB write occurs during the same clock cycle that the first write transfer ends on the memory bus.

2.14.2 Severity

High

A timing violation and possible data corruption may occur during back to back write transfers.

2.14.3 Symptoms

The chip select does not have a standard one cycle turnaround period, so the write enable signal will incorrectly only be deasserted for half a HCLK cycle before the chip select changes, and asserted on the same edge as the chip select. This may lead to a signal hold timing violation on the memory device, depending on the HCLK frequency and the memory timing requirements.

2.14.4 Short Term Workaround

This issue will not occur if the second AHB memory write transfer is started in a different cycle to that which the first write transfer ends on the memory bus.

If the second memory write is performed before or after the memory transfer finishes, the issue does not occur.

2.14.5 Affected Items

RTL.

2.14.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.15 SMCDATAOUT Is Generated One Cycle Late For Memory Writes With AHB Size (HSIZE) Greater Than The Memory Size (MSIZE)

2.15.1 Summary

For a memory write transfer with AHB size (HSIZE) greater than the memory size (MSIZE), SMCDATAOUT is incorrectly generated one cycle late for all memory transfers after the first write.

2.15.2 Severity

High

This issue will not occur if memory is used which is the same width as the AHB, or if only byte or halfword writes are performed to byte or halfword memories.

2.15.3 Symptoms

The overall transfer time is correct, but the first data write will be extended by one cycle, and the last data write (second or 4th write depending on the HSIZE and MSIZE settings) will be one cycle shorter, with all transfers in between shifted by one cycle. This may cause a data hold timing violation on the memory device for the last write transfer.

This issue will only occur for the following size settings:

- HSIZE = 32, MSIZE = 16
- HSIZE = 32, MSIZE = 8
- HSIZE = 16, MSIZE = 8

where a single AHB write will generate multiple writes to memory.

2.15.4 Short Term Workaround

A software workaround for this issue is to increase the WST2 timing value by one, which will make the last memory write the correct length, but will also cause all other writes in the transfer to be increased by one cycle.

2.15.5 Affected Items

RTL.

2.15.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.16 Issues With AHB WRAP4 Transfers and End of Burst Internal Signals

2.16.1 Summary

AHB WRAP4 transfers do not correctly assert the internal signal indicating the end of the burst (SmcEIB/NextBMlenEnd). This is not affected by the AHB transfer size or the memory size.

2.16.2 Severity

Medium

This issue does not occur if the AHB master never performs WRAP4 transfers.

2.16.3 Symptoms

This may lead to subsequent memory transfers being generated incorrectly.

2.16.4 Short Term Workaround

This issue will only occur if the AHB master generates WRAP4 burst transfers – all other AHB burst transfers will function correctly, including WRAP8 and WRAP16.

2.16.5 Affected Items

RTL

2.16.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.17 Clarification of WSTOEN and WSTWEN Programmed Values

2.17.1 Summary

For operation as specified in the Technical Reference Manual the WSTOEN programmed value must be less than or equal to WST1 and WSTWEN must be less than or equal to WST2.

2.17.2 Severity

Low

This issue will not occur if the software ensures the timing values, as specified in the Technical Reference Manual, are programmed in the timing registers.

2.17.3 Symptoms

If software incorrectly programs these timing registers, the transfers using these timing values may be generated incorrectly.

2.17.4 Short Term Workaround

Ensure that these registers are programmed as specified in the Technical Reference Manual.

2.17.5 Affected Items

RTL.

2.17.6 Proposed Corrective action

The r1p2 release contains a hardware fix to ensure that the register values are valid, by loading the WST1/2 value into the read/write enable counters respectively if the WSTOEN/WEN values are programmed greater than the WST1/2 values. This is a feature enhancement to match other current PrimeCell memory controllers and is not a bug fix.

2.18 Transfer State Machine Signal Not Deasserted at End of Burst Mode Read

2.18.1 Summary

If a burst mode memory read (BM=1) is followed by a write, the internal burst mode indicator signal (SmcTSM/BMRdTrans) is not deasserted at the end of the burst mode read.

2.18.2 Severity

High

Incorrect timing values may be used for read some transfers, causing read data to be generated incorrectly.

2.18.3 Symptoms

A read transfer following the write which is not to burst mode memory will incorrectly generate the transfer as if BM=1 for the current chip select. This means the memory transfer will be generated with the first read using WST1 timing, and any subsequent reads using WST2 timing.

The correct operation would be to use WST1 timing for all of the memory accesses.

2.18.4 Short Term Workaround

This issue does not occur if no burst mode (BM=1) memories are used in the system, or if all burst mode reads are followed by a read to non-burst mode memory (BM=0).

Also, if WST1=WST2 for the second read transfer, the timing of the memory accesses will not be affected.

2.18.5 Affected Items

RTL.

2.18.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

2.19 Burst Mode Indicator Incorrectly Deasserted At The End of The First Read of Two Read Transfers

2.19.1 Summary

If two read transfers are performed to burst mode memories (BM=1), the internal burst mode indicator will be deasserted at the end of the first read, and the second read will not use the burst mode timing (ie WST1 is used for all read transfers, instead of WST1 for the first read and WST2 for all subsequent reads).

2.19.2 Severity

Low

This issue only affects the performance of the second read transfer as burst mode timing is not used – the data returned will be correct.

2.19.3 Symptoms

As summary.

2.19.4 Short Term Workaround

This issue does not occur if no Burst Mode (BM=1) memories are used in the system.

2.19.5 Affected Items

RTL.

2.19.6 Proposed Corrective action

The issue has been fixed in version r1p2-00ltd0.

3 PL092 RELEASE R1P2 - ERRATA DETAILS

3.1 Booting is not possible from 16/32-bit devices with byte lane inputs on chip select 7

ARM reference D/E 222909

3.1.1 Summary

Tie-off inputs are provided to set the size of the memory connected to chip select 7, but no tie-off is provided to set the RBLE bit of the bank control register for chip select 7, which by default is set for 8-bit devices.

3.1.2 Severity

Low

Booting possible from 8-bit devices

3.1.3 Symptoms

Unable to read from 16-bit or 32-bit memory devices with byte lane inputs on chip select 7 immediately after reset. The nSMBLS outputs are only driven during write transfers when the RBLE bit of the bank control register is set to zero (8-bit device mode).

3.1.4 Short Term Workaround

Use 8-bit memory devices for booting, or 16/32-bit devices without byte lane inputs.

3.1.5 Affected Items

RTL

3.1.6 Proposed Corrective action

Add a tie-off input for the chip select 7 RBLE bit to allow configuration along with the memory size.

The issue has been fixed in version r1p3-00ltd0.

3.2 Issues Relating to SMWAIT Signal

ARM reference D/E 247412

3.2.1 Summary

If a number of transfers are performed to a number of different chip selects with different external wait enabled settlings (selecting the use of the SMWAIT input to control the timing of the transfer instead of the WST1 and WST2 registers) or to a single chip select with the external wait enabled, the turnaround time between the transfers may be incorrectly generated (the counter is loaded with the programmed turnaround value, but is then cleared in the following cycle, and the SMC may lock up by holding HREADYOUT LOW.

3.2.2 Severity

Medium

This issue will only occur if the SMWAIT external wait input is used to control the timing of the transfers on any chip select.

3.2.3 Symptoms

As summary.

3.2.4 Short Term Workaround

If possible, do not use the SMWAIT signal for any transfer, but use only the WST1 and WST2 registers to control the timing of the transfers.

3.2.5 Affected Items

RTL.

3.2.6 Proposed Corrective action

The issue has been fixed in version r1p3-00ltd0.

3.3 Issues With DBI/EBI

ARM reference D/E 247410

3.3.1 Summary

1) The SMC may incorrectly drive the external bus even when the grant input is not asserted to indicate that the SMC has control of the external memory bus. The grant may be from either the internal DBI or the external EBI, depending on the setting of the EXTBUSMUX input. This means that some transfers may be performed where the control signals are driven by the SMC and the address and data signals are driven by the device using either the MC port or the currently granted port of the EBI.

2) If the external memory bus grant is deasserted during memory accesses, subsequent transfers may be incorrectly generated. The grant may be from either the internal DBI or the external EBI, depending on the setting of the EXTBUSMUX input. Some cases may cause HREADYOUT to stay LOW locking up the AHB, and others may cause data transfers to be performed incorrectly, using the control signals from the SMC and address and data from the device which is granted control of the bus.

3.3.2 Severity

Medium

Both issues will only occur if another device is used to drive the external memory interface through the DBI or the EBI. If the SMC is the only device using the external bus then it will always be granted control of the external bus.

3.3.3 Symptoms

As summary

3.3.4 Short Term Workaround

If possible the EBI should not be used and the SMC should be the only device driving the external interface.

3.3.5 Affected Items

RTL.

3.3.6 Proposed Corrective action

This issue has been fixed in release r1p3-00ltd0

3.4 Double assignment to same signal in SmcAhbif.vhd

ARM reference D/E 277559

3.4.1 Summary

In the file SmcAhbif.vhd in the vhdl/rtl_source directory the waiten output signal is assigned the same value twice.

3.4.2 Severity

Medium - can be corrected

3.4.3 Symptoms

May get a compilation or synthesis error eg. In Synplicity Pro

3.4.4 Short Term Workaround

Line 3310 in SmcAhbIf.vhd can be removed

3.4.5 Affected Items

VHDL only – file SmcAhbIf.vhd

3.4.6 Proposed Corrective action

This issue has been fixed in release r1p3-00ltd0

4 PL092 RELEASE R1P3 - ERRATA DETAILS

4.1 SmcTM30 Timing Check Incorrect and Timing Violations README Missing

ARM reference D/E 288963

4.1.1 Summary

SmcTM30 is not checking the right timing values at the right times.

SmcTM30 checks the timing of the first access in a burst, but warnings are reported during the 2nd and 3rd accesses of a burst read (around 160ms for Smc1) but the SmcTrickMem is not checking the right timing values at the right times, where as it should have done SmcTM31 or SmcTM32 checks.

The timing violations README of the violations that occur in some simulations is missing.

4.1.2 Severity

Low

This is a trickbox error rather than an RTL functionality error.

4.1.3 Symptoms

There are violations that occur in Smc1 test that are not documented anywhere.

The SmcTM30 timing check errors are being generated on transfers which are being performed with the correct timing based on the programmed WST1/2 values of F/9 respectively, so the timing warnings should not occur.

This violation is reported by the SmcTrickMem block for burst mode reads with HSIZE is 32 bits and MSIZE is 32 bits, or HSIZE is 16 bits and MSIZE is 16 bits.

During this burst transfer, the state machine of SmcTrickMem moves to INITBURSTREAD state and remains there for subsequent transfers, resulting in SmcTM30 checks instead of SmcTM31.

The reason for this is that even though the memory width is 32 bits, the quad boundary which is checked to move to BURSTREAD state from INITBURSTREAD state is wrongly taken to be that of an 8-bit transfer. The quad boundary used for checking needs to be updated according to the memory width programmed.

4.1.4 Short Term Workaround

The following trick memory violations can be ignored :

1. SmcTM32: Access time violation for Read
This violation can be ignored since it is flagged when an SMC prefetch read is aborted at the end of a burst.

2. SmcTM30: Access time violation for Read
This violation can be ignored since it is flagged when an SMC prefetch read is aborted at the end of a burst.

3. SmcTM27: Maximum time limit for the burst read access has exceeded
This violation can be ignored since a prefetch read is aborted and then followed by a fresh read to the same address giving the impression that it is a single read. Since this read takes the burst read access time + the prefetch abort time of 2 clocks, the SmcTrickMem indicates that the access time has been exceeded.

4. SmcTM31: Maximum time limit for the burst read access has exceeded
This violation can be ignored. This is due to nSMOEN being deasserted after one read from memory for HSIZE > MSIZE and being asserted after the AHB has finished reading from the internal buffer.

5. SmcTM33: Maximum time limit for the burst read access has exceeded
This violation can be ignored. Here a prefetch read with zero access time is followed by a new burst read starting at the next address. Therefore the RTL does a normal mode read while the trick memory expects a burst mode read.

6. SmcTM25: Maximum time limit for the Burst mode read access has exceeded

This violation can be ignored since it is flagged when an SMC prefetch read is aborted at the end of a burst

7. SmcTM35 : Maximum time limit for the Burst mode read access has exceeded This violation can be ignored since it is flagged when an SMC prefetch read is aborted at the end of a burst.

4.1.5 Affected Items

SmcTrMemRdWrCtl.v and SmcTrMemRdWrCtl.vhd in the verification testbench.

4.1.6 Proposed Corrective action

These have been documented in the r1p3-01ltd0 release.

4.2 SmcState Machine Locks in ST_BUFWR State

ARM reference D/E 303810

4.2.1 Summary

When using the EBI to control access to memory, with the EXTBUSMUX input tied HIGH, the SMC state machine gets locked in the ST_BUFWR state, despite SMBUSGNT=1 and BufWrOver=1. This is due to the fact that one of the conditions for transitioning into another state is that SMBUSREQ is also asserted.

4.2.2 Severity

High – however, the scenario for this to occur is rare.

4.2.3 Symptoms

The scenarios that would cause this issue to occur are:

When using the EBI to control access to memory, with the EXTBUSMUX input tied HIGH, HREADYOUT is pulled LOW with the SMC remaining in the BUFWR state. This happens when both the following conditions occur:

1. On the AHB, an NSEQ-SEQ-IDLE or NSEQ-SEQ-NSEQ transfer sequence occurs, with a transfer size less than the width of the memory.
2. The EBI grants the SMC with the minimum possible delay.

A stale grant is the reason for the lock up occurring, when the SMC is granted but is not requesting grant. This happens because there is a single cycle latency between SMBUSGNT and SMBUSREQ.

4.2.4 Short Term Workaround

In the DBI block, the EBI grant passed to the SMC is gated with the EBI request from the SMC. This ensures that it is only granted when it is requesting control of the external memory bus.

In DBI.v

In the Wire Declarations:

```
// ---------------------------------------------------------------------------
// Wire declarations
// ---------------------------------------------------------------------------
```

add:

```
wire       SMBUSGNTEBIInt;
// SMBUSGNTEBI qualified with SMBUSREQ
```

then in the Main part of the code at:

```
// ---------------------------------------------------------------------------
// Multiplexer to choose between the DBI and the external EbiSdram for bus
// arbitration
// ---------------------------------------------------------------------------
```

add:

```
assign SMBUSGNTEBIInt   = SMBUSGNTEBI && SMBUSREQ;
```

and also change line 251 from:

```
assign SMBUSGNT         = (EXTBUSMUX == 1'b1) ? SMBUSGNTEBI : SMBUSGNTdbi;
```

to:

```
assign SMBUSGNT         = (EXTBUSMUX == 1'b1) ? SMBUSGNTEBIInt : SMBUSGNTdbi;
```

Similarly in DBI.vhd:

In the signal Declarations:

```
-- ---------------------------------------------------------------------------
-- Signal declarations
-- ---------------------------------------------------------------------------
```

add:

```
signal SMBUSGNTEBIInt  : std_logic;
-- SMBUSGNTEBI qualified with SMBUSREQ
```

then in the Main part of the code at:

```
-- ---------------------------------------------------------------------------
-- Multiplexer to choose between the DBI and the external EbiSdram for bus
-- arbitration
-- ---------------------------------------------------------------------------
```

add:

```
SMBUSGNTEBIInt   <= SMBUSGNTEBI and SMBUSREQ;
```

and change line 218 from:

```
SMBUSGNT          <= SMBUSGNTEBI  when (EXTBUSMUX = '1')
                  else
                     SMBUSGNTdbi;
```

to:

```
SMBUSGNT          <= SMBUSGNTEBIInt  when (EXTBUSMUX = '1')
                  else
                     SMBUSGNTdbi;
```

4.2.5 Affected Items

DBI.v/vhd

4.2.6 Proposed Corrective action

This issue has been fixed in release r1p3-01ltd0

5 PL092 RELEASE R1P3-01LTD0 - ERRATA DETAILS

At the time of release, there are no known issues with PL092 r1p3-01ltd0